

UCE401: HYDROLOGY AND GROUND WATER				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: The overall objective of this course is to acquaint students to the engineering aspects of water science. The students will learn different pathways of water movement, estimation and analysis of various components of pathways and its applications in designing various water related projects				
Introduction: Hydrologic cycle, Scope and Applications Precipitation: Measurement by rain gauge and other methods, estimation of missing data, consistency of records, optimum number of rain gauge station, mean precipitation, presentation and analysis of rainfall data Abstractions from Precipitation: Evaporation, factors affecting evaporation, measurement, infiltration, factors affecting infiltration, measurement, infiltration indices Run off: Run-off estimation, rainfall-runoff correlation, flow duration curve, hydrographs, base flow separation, unit hydrographs and its application, distribution graph, synthetic unit hydrograph Stream flow measurement: Velocity measurement: floats, velocity rods, current meters, discharge computation: velocity area method, moving boat method, slope area method, stage discharge curve, notches, weirs, venturiflume, standing wave flume, free overfall Floods Frequency analysis: Peak flood estimation, methods of frequency analysis, flood routing Ground Water Hydraulics: Type of aquifers, aquifer constants, Darcy's law, Steady flow towards fully penetrating well, Equation of motion and its applications to ground water flow problems, introduction to the use of distributed groundwater				
Course Learning Objectives (CLO) Upon the completion of this course, the students will be able to: 1. Perform analysis on precipitation, evaporation and infiltration data for various applications 2. Estimate runoff and discharge generated from watershed and/or stream by using various techniques 3. Apply the principles of flood frequency analysis and flood routing to forecast floods 4. Apply hydraulic principles of groundwater flow in different geological formations				
Text Books: 1. Subramanya, K., Engineering Hydrology, Tata McGraw-Hill Publication (2015) 2. Reddy, Jaya Rami., A textbook of Hydrology, University Science Press, New Delhi (2014) 3. Raghunath, H. M., Hydrology: Principles, analysis and Design, New Age International				

Publishers (2015)

Reference Books:

1. Chow, V.T., Maidment, D.R., and Mays, L.W., Applied Hydrology, Mc-Graw-Hill International Editions, New York (2006)
2. Misstear, B., Banks, D., and Clark, L., Water Wells and Boreholes, John Wiley & Sons Ltd, UK (2013)
3. Shaw, E.M., Beven, K.J., Chappell, N.A., and Lamb, R., Hydrology in Practice, Spon Press, New York (2014)
4. Viessman, W. and Lewis, G.L., Introduction to Hydrology, Prentice Hall of India Pvt Limited, New Delhi (2012)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25